

Moussa Hamzi

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QUALIFICATIONS

- Computer Architecture, Computer Systems Design, VHDL, RTL
- Semiconductor Fabrication
- 2.5D Heterogenous Integration
- Strong Interpersonal and Communication Skills, Adaptability, and Teamwork and Collaboration

EDUCATION

Master of Science, Electrical Engineering, *University of Dayton, School of Engr.*, Dayton, OH 05/2028

- Relevant Courses: Computer Design, Contemporary Digital Systems

Bachelor of Electrical Engineering, *University of Dayton, School of Engr.*, Dayton, OH 05/2026

- Relevant Courses: Discrete Signals and Systems, PCB Design and Manufacturing, Computer Systems Engineering, Electronic Devices, Intro to Semiconductor Manufacturing, Data Analytics Fundamentals

CERTIFICATIONS

Altium Education PCB Design Course 04/2026

Intel Oasis Rapid Certification 05/2025

PROJECTS

GPU Core – *University of Dayton*, Dayton, OH 09/2026

- Designed a 128-lane SIMD GPU execution core in VHDL with 16 execution units and 4 independent schedulers
- Implemented round-robin scheduling, dependency tracking, and register writeback
- Built independent program sequencing and register storage for multiple resident groups per SIMD engine

Vision Walker System – *University of Dayton*, Dayton, OH 10/2025 – 05/2026

- Led senior design team in building a real-time assistive navigation system for blind and deaf individuals
- Built on Jetson Orin Nano, uses stereo cameras and a time-of-flight sensor for 3D spatial awareness

Pipelined MIPS Simulator – *University of Dayton*, Dayton, OH 09/2025 – 11/2025

- Converted an unpipelined MIPS processor simulator written in C into a 5-stage pipelined simulator
- Implemented core pipeline behavior incl. data forwarding, stalling, branch/jump handling, and write-back

RESEARCH EXPERIENCE

Memristor Crossbar Application – *University of Dayton*, Dayton, OH 10/2025 – Present

- Construction of a 4x4 Memristor Crossbar using Knowm Memristors through breadboarding
- Collecting measurement data and comparing performance, efficiency, with SPICE simulations
- Crossbar completes mathematical operations in analog and stores data digitally

Silicon 2.5D Packaging Research – *University of Dayton*, Dayton, OH 10/2025 – 05/2026

- Packaged silicon dies using Nano Dimension Dragonfly IV PCB 3D Printer
- Created dielectric package, with die glued into the base, and vias, silver ink traces and test pads printed above

AI Band-gap Energy Prediction – *University of Dayton*, Dayton, OH 09/2024 – 12/2024

- Predicted Band-gap Energy of TMDC's by converting Euclidean coordinates to canonical coordinates using Python
- Used AI algorithms SVR and SVC, with a mean squared error (MSE) of 0.00695 and R-squared value of 0.974

EXPERIENCE

Teaching Assistant – *School of Engr., ECE Department*, UD, Dayton, OH 09/2026 – Present

- Instructing ECE 215L classes, teaching students how to implement basic digital circuits on FPGA boards

Engineering Lab Assistant – *School of Engr., Lab Management Office*, UD, Dayton, OH 05/2024 – Present

- Support lab management team in daily needs, using technological and non-technological solutions
- Created student training records management and sign in app for University's Makerspace
- Troubleshooted oscilloscopes, power supplies, multimeters, and function generators
- Trained over 50 high school students in soldering during single day events